Comparison of surds:

1. Increasing Power — $\sqrt[3]{25}$ and $\sqrt{8}$
2. Decreasing Power — 5^{54} and 11^{36}
3. Squaring both sides — $6+\sqrt{5}$ and $5+\sqrt{11}$
4. Observing pattern

Must memorize:

- $\sqrt{2} = 1.414$
- $\sqrt{3} = 1.732$
- $\sqrt{5} = 2.236$

Which of the following is greatest?

a) $\sqrt{2}$ b) $\sqrt[3]{3}$ c) $\sqrt[4]{4}$ d) $\sqrt[6]{6}$

Find the smallest among $2^{\frac{1}{12}}, 3^{\frac{1}{18}}, 4^{\frac{1}{24}}, 6^{\frac{1}{36}}, 12^{\frac{1}{72}}$ is

a) $2^{\frac{1}{12}}$ b) $3^{\frac{1}{18}}$ c) $4^{\frac{1}{24}}$ d) $6^{\frac{1}{36}}$ e) $12^{\frac{1}{72}}$

The greatest one of $\sqrt{2}$, $\sqrt[3]{3}$, $\sqrt[6]{6}$, $\sqrt[5]{5}$ is

$\sqrt{}$ 3, 6, 5

a) $\sqrt{2}$ b) $\sqrt[3]{3}$ c) $\sqrt[6]{6}$ d) $\sqrt[5]{5}$

The greatest among the numbers $\sqrt[2]{8}, \sqrt[4]{13}, \sqrt[5]{16}, \sqrt[10]{41}$ is :

8, 13, 16,

a) $\sqrt[4]{13}$ b) $\sqrt[5]{16}$ c) $\sqrt[10]{41}$ d) $\sqrt[2]{8}$

The greatest number among $3^{50}, 4^{40}, 5^{30}$ and 15^{20} is

a) 3 b) 4 c) 5 d) 15

The greatest among $2^{550}, 3^{300}, 5^{250}, 6^{200}$ is

a) 2 b) 3 c) 5 d) 6^{200}

Find the smallest among the following

a) $1 + 3\sqrt{2}$ b) $2 + \sqrt{10}$ c) $3 + \sqrt{5}$ d) $4 + \sqrt{3}$

Which of the following statement(s) is/are true?

I. $\sqrt{11} + \sqrt{7} < \sqrt{10} + \sqrt{8}$

II. $\sqrt{17} + \sqrt{11} > \sqrt{15} + \sqrt{13}$

a) Only I b) Only II

c) Both I and II d) Neither I nor II

Which of the following statement(s) is/are TRUE?

I. $\sqrt{5} + \sqrt{5} > \sqrt{7} + \sqrt{3}$

II. $\sqrt{6} + \sqrt{7} > \sqrt{8} + \sqrt{5}$

III. $\sqrt{3} + \sqrt{9} > \sqrt{6} + \sqrt{6}$

a) Only I b) Only I and II c) Only II and III d) Only I and III

Arrange the following expressions in descending order:

P) $2\sqrt{2} + \sqrt{7}$

Q) $3 + \sqrt{6}$

R) $\sqrt{10} + \sqrt{5}$

S) $\sqrt{13} + \sqrt{2}$

a) R $> Q > P > S$

b) $P > Q > S > R$

c) $P > Q > R > S$

d) R $> P > S > Q$

Arrange the following expressions in descending order:

P) $\sqrt{22} - \sqrt{12}$

Q) $\sqrt{93} - \sqrt{83}$

R) $\sqrt{15} - \sqrt{5}$

S) $\sqrt{37} - \sqrt{27}$

a) $R > S > P > Q$

b) $S > P > R > Q$

c) $P > R > S > Q$

d) $R > P > S > Q$

Arrange the following expressions in descending order:

P) $\sqrt{17} - \sqrt{15}$

Q) $2\sqrt{3} - \sqrt{10}$

R) $3 - \sqrt{7}$

S) $\sqrt{19} - \sqrt{17}$

a) $R > Q > P > S$

b) $S > P > R > Q$

c) $P > R > S > Q$

d) $R > P > S > Q$

Which of the following is True

a) $\sqrt[3]{11} > \sqrt{7} > \sqrt[4]{45}$

b) $\sqrt{7} > \sqrt[3]{11} > \sqrt[4]{45}$

c) $\sqrt{7} > \sqrt[4]{45} > \sqrt[3]{11}$

d) $\sqrt[4]{45} > \sqrt{7} > \sqrt[3]{11}$

Which of the following statements(s) is/are true?

1. $(65)^{\frac{1}{6}} > (17)^{\frac{1}{4}} > (12)^{\frac{1}{3}}$
2. $(17)^{\frac{1}{4}} > (65)^{\frac{1}{6}} > (12)^{\frac{1}{3}}$
3. $(12)^{\frac{1}{3}} > (17)^{\frac{1}{4}} > (65)^{\frac{1}{6}}$

a) only 1
b) only 2
c) only 3
d) none of these

Which of the following is true?

I. $\frac{1}{\sqrt[3]{12}} > \frac{1}{\sqrt[4]{29}} > \frac{1}{\sqrt{5}}$

II. $\frac{1}{\sqrt[4]{29}} > \frac{1}{\sqrt[3]{12}} > \frac{1}{\sqrt{5}}$

III. $\frac{1}{\sqrt{5}} > \frac{1}{\sqrt[3]{12}} > \frac{1}{\sqrt[4]{29}}$

IV. $\frac{1}{\sqrt{5}} > \frac{1}{\sqrt[4]{29}} > \frac{1}{\sqrt[3]{12}}$

a) Only I b) Only II c) Only III d) Only IV